A Fuzzy TOPSIS Framework for Prioritising Pedestrian Crossing Safety and Accessibility Improvements

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Abstract

This report asks whether adding essential-destination access (E1) changes the leading group beyond a four-criterion baseline. In the 28-candidate simulation, the five-criterion model gives a tied top score of 0.6839; the top-three overlap with the four-criterion model is 0.667, and 1,000 bounded weight draws have mean top-three overlap 1.000. The pedestrian-demand/E1 central-score Spearman correlation is 0.80, indicating substantial constructed-proxy overlap. These are synthetic calculations informed by public map geometry—not measured conditions, verified crossings, or an empirical safety ranking. The older four-row example remains a separate teaching illustration.

Keywords: pedestrian crossings; accessibility; fuzzy TOPSIS; synthetic data; sensitivity analysis

I Introduction

A transparent multi-criteria ranking can organise review when demand, road context, accessibility and control evidence are incomplete, but its meaning depends on its inputs and candidate set. This project ranks hypothetical improvement packages for earlier review—not danger, crash prevention, or a budget-optimal portfolio. Its contribution is a reproducible context-informed synthetic dataset: unlike independently random alternatives, candidate selection follows saved road geometry and shared-node junction topology, while road hierarchy and mapped destinations inform explicit proxy formulas. This creates traceable intermediate evidence and assumptions, not stronger empirical observations; modeled deficiencies remain prior-based and every score remains synthetic. Feasibility is an external engineering and jurisdictional gate.

II Related Work

Structured crossing checklists [1], pedestrian-facility prioritisation [2], and fuzzy gap-acceptance analysis [3] motivate documented evidence and decision support, but their settings and outcomes do not validate these inputs. Chen [4] informs the TOPSIS lineage; fuzzy conversion research [5] motivates explicit sensitivity. Kerala's road-safety vision [6] and a midblock study from four other Indian cities [7] provide context only. OSM [8] supplies the mapped layer; Kerala district charts [9] are not score inputs. The declared calculations below are project adaptations, not an unchanged reproduction of an earlier formulation.

Study	Data and approach	Relevance and transfer limit
Battistini et al. [1]	Checklists at ten unsignalized crossings on two Bologna roads; decision support for intervention priorities.	Supports structured inspection; Italian site ratings are not local observations or Indian compliance thresholds.
Majumdar et al. [2]	Pedestrian satisfaction and TOPSIS prioritisation of sidewalk/crosswalk attributes in an Indian context.	Supports pedestrian-focused criteria; ranking infrastructure attributes differs from ranking crossing packages.
Alver and Katanalp [3]	Integrated AHP and fuzzy logic for gap acceptance at semicontrolled midblock crossings.	Illustrates fuzzy transport modelling; behavioral analysis does not validate our synthetic deficiency scale.

III Study Scope and Data

The generator reads the saved data/raw/osm_context.json extract and data/raw/osm_metadata.json metadata. The extract contains 431 elements (286 accepted motor-road ways; 91 deduplicated destinations). From the mapped center (9.7554364° N, 76.6495350° E), a deterministic 3 km search, ≥80 m spacing, and map-evidence priority selected 28 coordinates: 27 road-junction candidates and one explicitly mapped crossing. All are unverified. Shared-node junctions are candidate locations, not proof of crossings; road ownership, field conditions, and map completeness are unknown.

Evidence chain: saved map geometry/tags and destination objects are mapped facts; formulas derive context indicators; deterministic seeded rules produce synthetic ordinal centers and ranges. Example SYN-OSM-716872373 is near mapped St Mary's Church (60 m), Alphonsa English Medium School Valavoor (83 m), Valavoor School (277 m), and a post office (286 m), all straight-line distances. Proximity does not show a walking route or access need. All centers are integer scores 1–5. Fuzzy widths are ±1.0 for low-confidence proxies and ±1.5 for very-low/unknown assumptions, clipped to scale bounds; these are not confidence intervals.

Criterion	Map/context input and generation rule	Interpretation
P1 Demand	Nearby mapped amenity intensity I, bus stop B, vehicle score; round_clip(1+1.05 ln(1+I) +0.35B+0.12(vehicle−2)+U(−0.65,0.65)).	Constructed demand proxy; no counts.
P2 Vehicle	Highest adjacent road class (motorway/trunk/primary 5; secondary 4; tertiary 3; residential/unclassified 2; service/living street 1); +0.5 for each satisfied condition: ≥3 mapped lanes, numeric maxspeed ≥60 km/h; U(−0.35,0.35).	Road context, not volume or operating speed.
P3 Access	Prior 3 + shared U(−0.45,0.45) + N(0,0.85); mapped wheelchair, kerb/sloped curb and tactile tags adjust it.	Unknown tags stay unknown; deficiency mostly assumed.
P4 Control	Prior 3 + shared variation +0.2(vehicle−3)+N(0,0.85); mapped signals/markings and lighting adjust it.	Not a visibility or condition audit.
E1 Destination	Strongest mapped destination proximity plus total intensity: round_clip(1+1.05M+0.25 ln(1+I)).	Straight-line proxy, not walking-network access.

Within 500 m, each destination contributes a(1−d/600), where d is great-circle distance in metres and a is the declared destination-type weight. I sums these contributions; M is their maximum (zero when none are mapped). B=1 if a mapped bus stop/platform lies within 250 m, otherwise 0. Schools/clinics/hospitals have a=3, universities/pharmacies/markets 2.5, bus stops 2, banks/post offices/community or worship facilities 1.5, and other shops 1. round_clip rounds to the nearest integer and clips to 1–5. N(0,0.85) denotes mean zero and standard deviation 0.85; all draws are seeded assumptions.

Scores/ranges are synthetic, not calibrated. For deficiency criteria, the prior is 3 and missing tags do not imply absence. Generator seed 20261024 differs from sensitivity seed 20261009. Exact feature weights, road-class constants, draw order, and source records are in reports/synthetic_data_methodology.md and data/source_inventory.csv. No counts, surveys, operating speeds, expert validation, or feasibility findings were invented.

IV Methodology

For alternative i and criterion j, represent the priority-oriented input as a triangular fuzzy number $\tilde{x}_{ij}=(l,c,u)$, where $l\le c\le u$. Components are normalized against shared input outer bounds $L_j=\min_i l_{ij}$ and $U_j=\max_i u_{ij}$: $z=(x-L_j)/(U_j-L_j)$. Non-discriminating criteria are removed and remaining equal weights renormalized. With five active criteria, each weight is 1/5. Weighted triangles are compared with theoretical positive ideals (w_j,w_j,w_j) and negative ideals (0,0,0). The positive ideal means greatest intervention priority, not the safest condition.

The vertex distance is $d(\tilde{a},\tilde{b})=\sqrt{((a_l-b_l)^2+(a_c-b_c)^2+(a_u-b_u)^2)/3}$. Criterion distances are summed: $D_i^+=\Sigma_j d(\tilde{v}_{ij},A_j^+)$, $D_i^-=\Sigma_j d(\tilde{v}_{ij},A_j^-)$; closeness is $C_i=D_i^-/(D_i^++D_i^-)$. Descending C ranks higher priority. This is a declared project adaptation: shared-range normalization, equal weights, weighted theoretical ideals, and summed distances. Fuzzy membership grades are not probabilities; ranges are uncalibrated stress-test representations.

Research question: does adding E1 change ordering while the four original criteria are retained? E1 is a 500 m straight-line proxy, not the field definition based on verified walking routes and access necessity. Demand and E1 share amenity context; their central-score Spearman correlation is 0.801, so double-counting is plausible. The crisp comparator min–max scales central values and averages active columns; it is not crisp TOPSIS. The comparison does not establish predictive accuracy.

Processing path and comparison design

Stage	Input → output	What enters model
Select candidates	osm_context.json + osm_metadata.json → mapped candidates/context	Coordinates, road/destination context
Generate synthetic rows	build_synthetic_dataset.py → processed/pedestrian_crossings_fuzzy_topsis.csv	Five synthetic criterion triples
Run CLI	--synthetic-input reads that CSV → scores/ranks	Criterion triples only; metadata fields validate provenance, not criteria
Evaluate/report	evaluate_report.py → comparisons, sensitivity, calculation trace	Same CSV; four- and five-criterion analyses

The CLI does not reread the OSM files or the field observation template: those inputs are upstream of the generated CSV. The model receives the five criterion triples; identifiers and provenance metadata do not become extra TOPSIS criteria.

V Implementation and Evaluation

scripts/build_synthetic_dataset.py generates the synthetic CSV from the preserved OSM extract; the existing CLI/model pipeline evaluates it. scripts/evaluate_report.py regenerates reports/evaluation/summary.json, seeded weight runs, uncertainty scenarios, and a per-criterion trace. This builder reads that summary, including the input SHA-256 and runtime versions; numerical results are not hardcoded. Tests assess reproducibility and arithmetic consistency, not empirical validity.

The recorded execution uses Python 3.14.8, NumPy 2.5.4 and pandas 2.3.3. Validation checks required metadata, unique IDs, coordinates, integer central scores and ordered finite bounds in 1–5; missing values are rejected rather than set to zero. The calculation is deterministic: randomness is confined to the documented synthetic generator and explicitly seeded sensitivity experiments. Appendix B reconstructs one complete five-criterion result.

VI Results

Rank	OSM node ID	Mapped context (hypothetical)	Score	4-criterion rank	Crisp rank
1	716872373	Pala - Valavoor - Uzhavoor Road	0.6839	1	1
1	764885861	Pala - Valavoor - Uzhavoor Road	0.6839	1	1
3	679043325	Valavoor - Chakkampuzha Road (Part)	0.6504	4	4
4	679049378	Valavoor - Chakkampuzha Road (Part)	0.6138	3	3
5	4778312540	Pala Vaikom Road	0.5829	6	9
6	9119950895	palackattumala-valavoor road	0.5808	7	8
7	679049494	Valvoor - Vellappura Road	0.5735	5	5
8	678584549	Pala - Valavoor - Uzhavoor Road	0.5587	10	16

Table 1 reports the leading eight of 28 alternatives under the five-criterion model. Candidate IDs are SYN-OSM- followed by the displayed node ID; full IDs appear in Appendix A. The score and ranks describe this synthetic candidate set only; shared-bound normalization means adding or removing alternatives can change results. The comparisons below use the same alternatives.

Comparison	Top-1 changed	Top-3 overlap	Spearman
base vs crisp	False	1.000	0.964
base vs extension	False	0.667	0.909

The leading two candidates exchange their demand and vehicle fuzzy triples while their other three triples match. Equal weights yield equal scores; weight changes expose that trade-off rather than establishing a unique best location. Adding E1 exchanges the third and fourth baseline positions, so two of the three selected leading alternatives remain shared. This illustrates a controlled extension effect, not a before/after safety improvement. Rounded scores should not be used to infer meaningful separation of near-tied candidates.

Sensitivity to assumptions

Weight sensitivity uses 1,000 seeded perturbations: each equal weight is multiplied by an independent U(0.8,1.2) factor and renormalized. These are exploratory scenarios, not stakeholder preference probabilities. Width sensitivity contracts triangles toward their centers by $\alpha \in \{0,0.5,1\}$ and recomputes normalization; it tests the full representation/scaling procedure, not fuzziness in isolation. Exact ties share minimum rank; top-choice frequencies use deterministic ID ordering, not probabilities.

The five-criterion top group is tied between SYN-OSM-716872373 and SYN-OSM-764885861 at 0.6839. Adding E1 gives top-three overlap 0.667 versus the four-criterion baseline (Spearman 0.909); this is an assumption comparison, not validation.

Across 1000 weight draws, mean top-three overlap with equal weights is 1.000 (minimum 1.000); mean Spearman rank correlation is 0.990 (minimum 0.964). These perturbations are scenarios, not probabilities of field priority.

Uncertainty-width scenarios (α, top-ranked ID(s), Spearman versus full width): 0.0: SYN-OSM-716872373, SYN-OSM-764885861, 0.977; 0.5: SYN-OSM-716872373, SYN-OSM-764885861, 0.985; 1.0: SYN-OSM-716872373, SYN-OSM-764885861, 1.000.

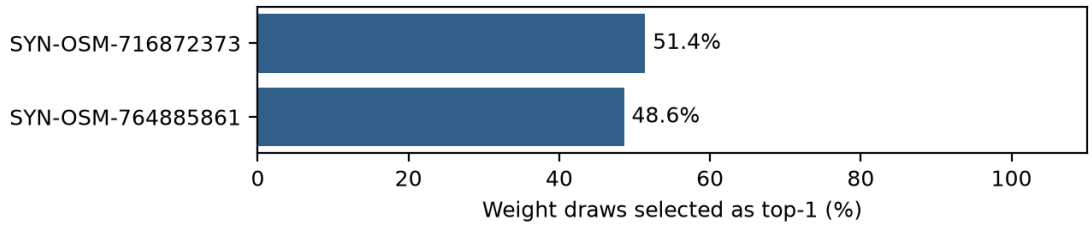


Figure 1. Only the 2 alternatives selected at least once are shown; the other 26 candidates had zero selections. Deterministic tie handling applies. These are experiment frequencies, not probabilities of need.

VII Discussion and Limitations

The experiment demonstrates computational behavior, not relative safety. A high closeness score means proximity to the modeled high-priority ideal within this candidate set. The public map extract supports candidate geometry and contextual proxies, but 27 junction candidates and the one mapped crossing remain unverified. In particular, road class is not traffic volume or operating speed; mapped tags do not certify current conditions; unknown attributes remain unknown. The destination proxy is not a verified route or access measure, and shared amenity inputs may count context twice.

Central-score demand/destination association is reported as a diagnostic, not evidence that criteria are empirically independent. The five-versus-four comparison tests the mathematical consequence of adding E1 at equal weight, not predictive accuracy or improved decisions. The crisp comparator changes representation and scaling. Sensitivity ranges are analyst-chosen, fuzzy bounds uncalibrated, and equal weights provisional. A stable or unstable rank under these scenarios cannot validate the assumptions.

Before operational use, collect matched pedestrian and vehicle counts, independently documented accessibility/control audits, current photographs, walking-route and destination verification, dates and disruptions, costs, jurisdiction, and feasibility. Preserve missingness and disagreements, then obtain independent review against predefined evidence. The model cannot establish code compliance, crash risk, treatment effect, or a budget-optimal portfolio. A low rank does not imply safety; a high rank does not quantify benefit.

VIII Conclusion

A reproducible five-criterion simulation demonstrates how a declared fuzzy TOPSIS adaptation can organize hypothetical crossing-improvement candidates and expose sensitivity to weighting and range assumptions. Its evidence-informed geography is not field evidence: public map facts, derived indicators, and modeled scores are kept distinct. Results must not be presented as validated site priorities. The report retains an older four-alternative example separately for teaching and provides reproducible artifacts for later replacement with verified observations.

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Appendix A. Complete synthetic five-criterion ranking

Rank	Alternative ID	Score	4-criterion rank	Crisp rank
1	SYN-OSM-716872373	0.683920	1	1
1	SYN-OSM-764885861	0.683920	1	1
3	SYN-OSM-679043325	0.650427	4	4
4	SYN-OSM-679049378	0.613844	3	3
5	SYN-OSM-4778312540	0.582940	6	9
6	SYN-OSM-9119950895	0.580800	7	8
7	SYN-OSM-679049494	0.573485	5	5
8	SYN-OSM-678584549	0.558692	10	16
9	SYN-OSM-2463032701	0.550469	11	15
10	SYN-OSM-718505044	0.532603	8	6
11	SYN-OSM-9878266128	0.495477	19	14
12	SYN-OSM-8361480469	0.492011	8	6
13	SYN-OSM-2937370792	0.467163	17	18
14	SYN-OSM-13132320792	0.462831	12	10
14	SYN-OSM-800653779	0.462831	12	10
14	SYN-OSM-9878266129	0.462831	22	20
17	SYN-OSM-13046000350	0.459742	23	21
18	SYN-OSM-9878266122	0.459456	18	17
19	SYN-OSM-2249909190	0.443076	12	10
19	SYN-OSM-5970919652	0.443076	12	10
21	SYN-OSM-764885749	0.427036	16	18
22	SYN-OSM-12716637337	0.388449	24	24
23	SYN-OSM-874214059	0.370067	20	21
23	SYN-OSM-8859297040	0.370067	20	21
25	SYN-OSM-12716637471	0.364076	28	28
26	SYN-OSM-12716637312	0.356273	25	25
27	SYN-OSM-13132320783	0.296566	25	25
28	SYN-OSM-2937370755	0.296566	25	25

Appendix B. Worked five-criterion calculation

Alternative SYN-OSM-716872373: raw and normalized triples, equal weight, weighted triple, and criterion-level vertex distances are shown below. Distances sum across criteria; C = D−/(D+ + D−).

Criterion	Raw (l,c,u)	Normalized	Weight	Weighted	d+	d−
ped	[3.0, 4.0, 5.0]	[0.5, 0.75, 1.0]	0.200	[0.1, 0.15, 0.2]	0.064550	0.155456
vehicle	[4.0, 5.0, 5.0]	[0.75, 1.0, 1.0]	0.200	[0.15, 0.2, 0.2]	0.028868	0.184842
access	[1.5, 3.0, 4.5]	[0.125, 0.5, 0.875]	0.200	[0.025, 0.1, 0.175]	0.117260	0.117260
deficiency	[2.5, 4.0, 5.0]	[0.375, 0.75, 1.0]	0.200	[0.075, 0.15, 0.2]	0.077728	0.150693
destination	[3.0, 4.0, 5.0]	[0.5, 0.75, 1.0]	0.200	[0.1, 0.15, 0.2]	0.064550	0.155456

Totals: D+=0.35295551, D−=0.76370815, C=0.68391959.

For the demand criterion, shared bounds are 1 and 5: (3,4,5) becomes (0.5,0.75,1), then weight 0.2 gives (0.1,0.15,0.2). Its positive-ideal distance is $\sqrt{[(0.1-0.2)^2+(0.15-0.2)^2+(0.2-0.2)^2)/3]} = 0.06454972$. Repeating for each criterion and summing gives the totals above; 0.76370815/(0.35295551+0.76370815) = 0.68391959. These calculations explain a synthetic result, not measured safety.

Appendix C. Separate four-alternative teaching illustration

The original hypothetical_demo.csv is a separate small teaching illustration; its invented rows are not part of the 28-candidate map-informed experiment. Its rankings are regenerated by the original demo pipeline and kept distinct.

ID	4-criterion score	4-criterion rank	Crisp rank	5-criterion score	5-criterion rank
HYP-A	0.595646	1	1	0.565904	2
HYP-B	0.561219	2	2	0.559114	3
HYP-C	0.505106	4	2	0.570867	1
HYP-D	0.559763	3	4	0.498248	4

The four criteria are pedestrian demand, vehicle exposure, accessibility deficiency, and control/visibility deficiency; the five-criterion extension adds essential-destination access. Unlike the main 1–5 dataset, the original example uses invented hourly rates and 0–3 audit ratings, normalized within its own four rows. Raw triples remain in data/hypothetical_demo.csv. HYP-C moves from fourth to first after E1 is added, illustrating rank reversal; this does not validate the larger experiment. Scores from the two candidate sets are not directly comparable.

Appendix D. Reproduction

From the repository root, generate the map-informed synthetic CSV with python scripts/build_synthetic_dataset.py; run the synthetic-input CLI using the documented command in README.md and its dedicated synthetic output directory; then run python scripts/build_report.py. The report builder invokes scripts/evaluate_report.py and writes reports/evaluation/summary.json plus reports/report_figures/weight_top1_sensitivity.png. The four-row demo remains separately available through python src/run_project.py --demo. Do not use either synthetic dataset as field input. See reports/synthetic_data_methodology.md and data/source_inventory.csv for exact inputs, provenance, formulas, and limitations.